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(54) **ELECTRET CAPSULE CIRCUIT**

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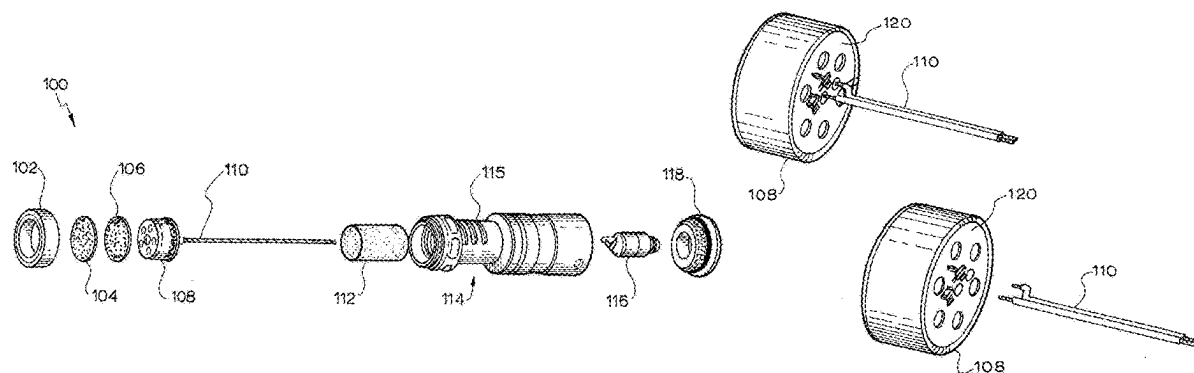
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(57)

**ABSTRACT**

The present invention relates generally to the field of electret capsule circuit, and more particularly to a circuit configuration of an impedance converter integrated in an electret capsule such as for use in condenser microphones. The electret capsule of a microphone may include a gate biasing field effect transistor (FET) to facilitate biasing of a low noise FET. Advantageously, the use of low noise FET in the electret capsule of a microphone provides for a reduced cost, while achieving lower self-noise.



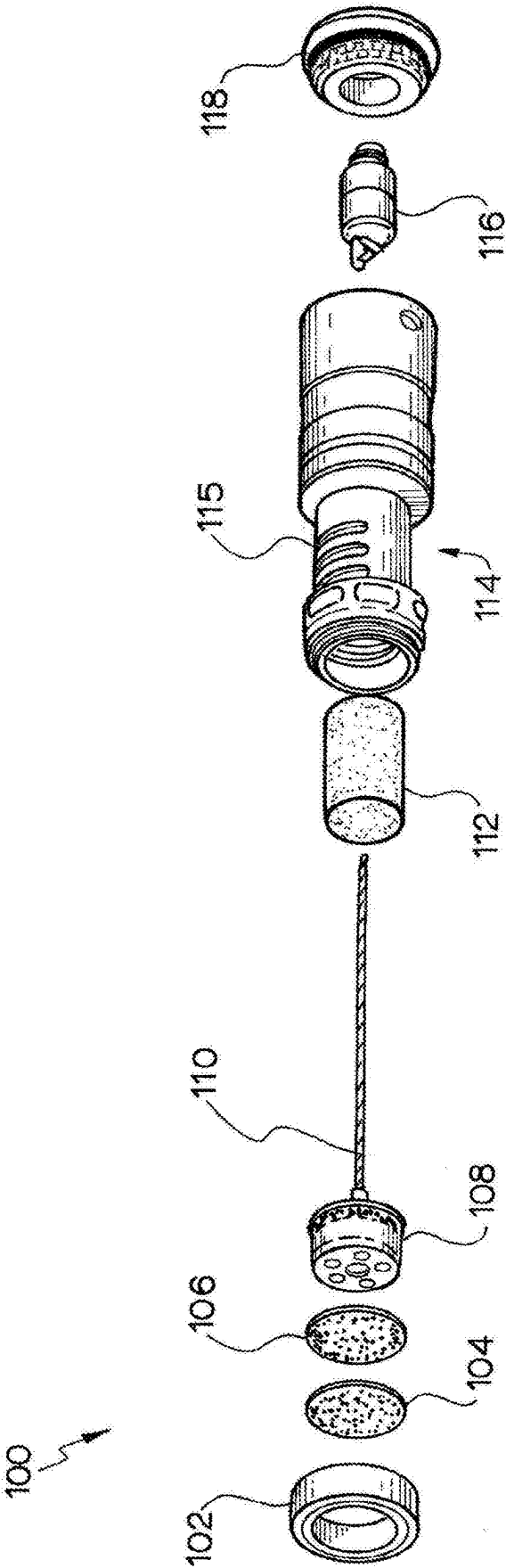


FIG. 1A

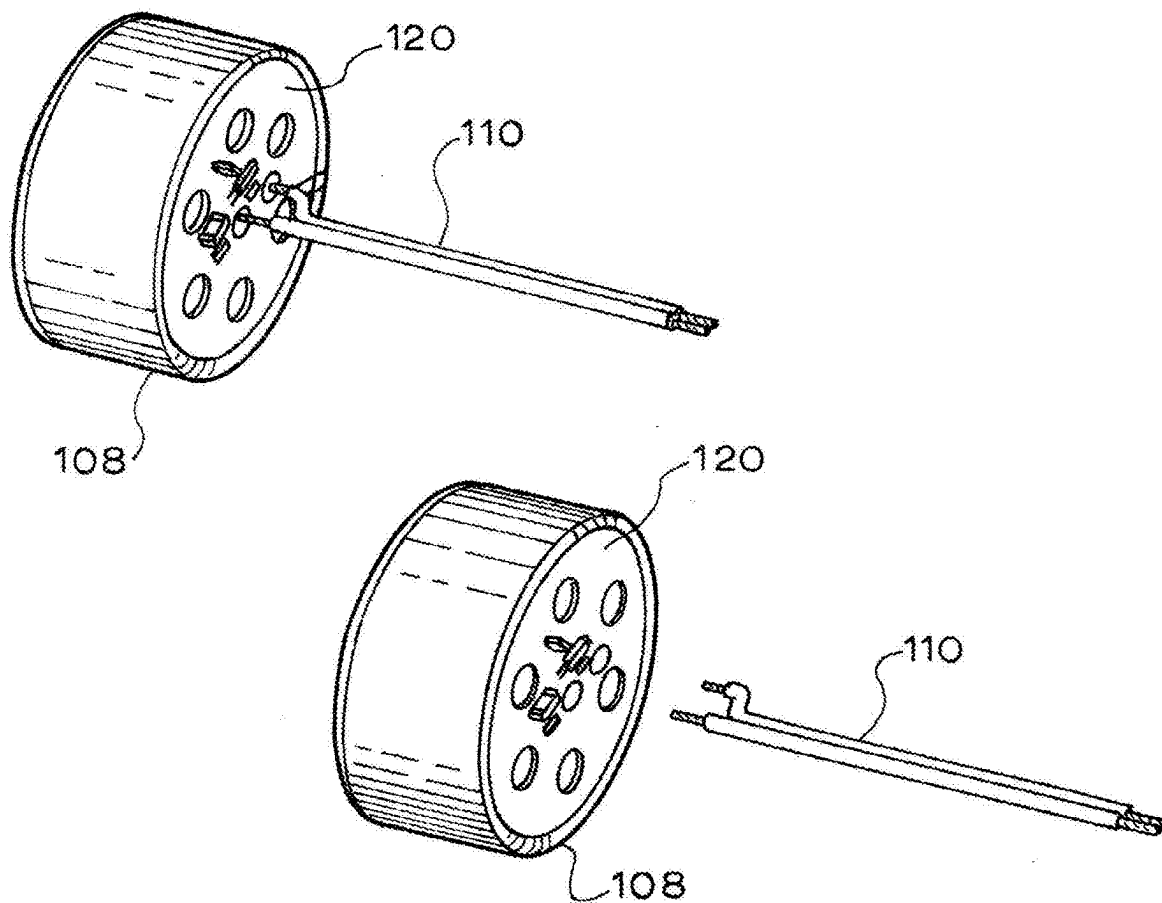


FIG. 1B

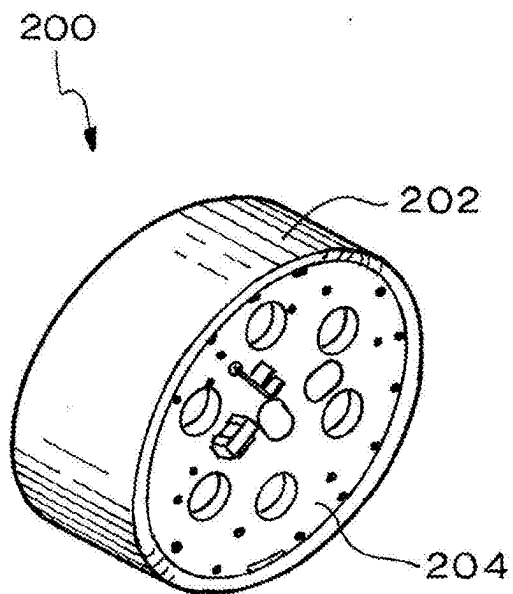


FIG. 2A

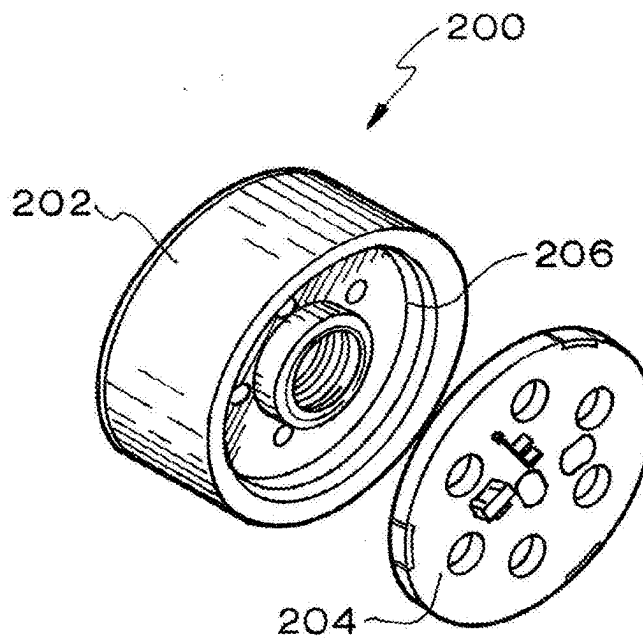


FIG. 2B

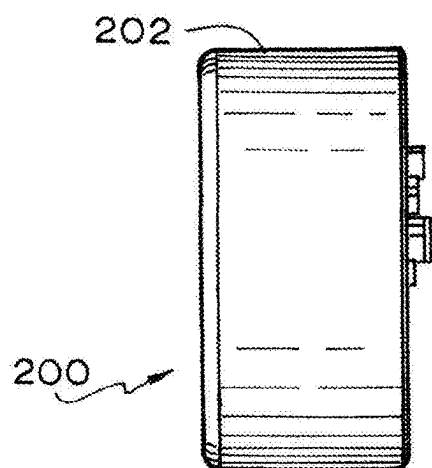


FIG. 2C

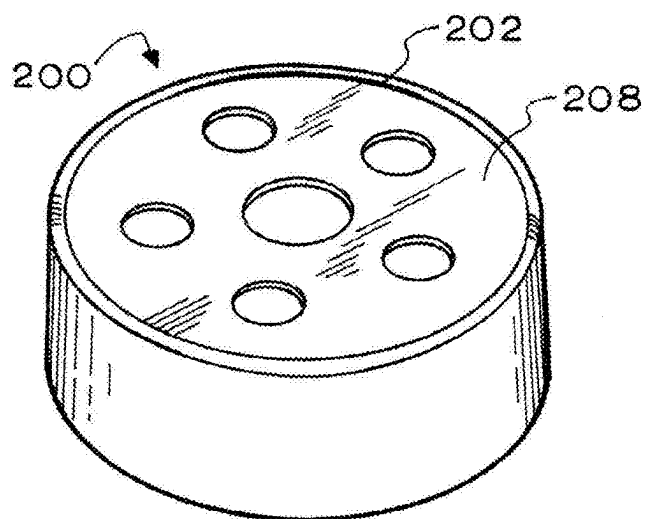


FIG. 2D

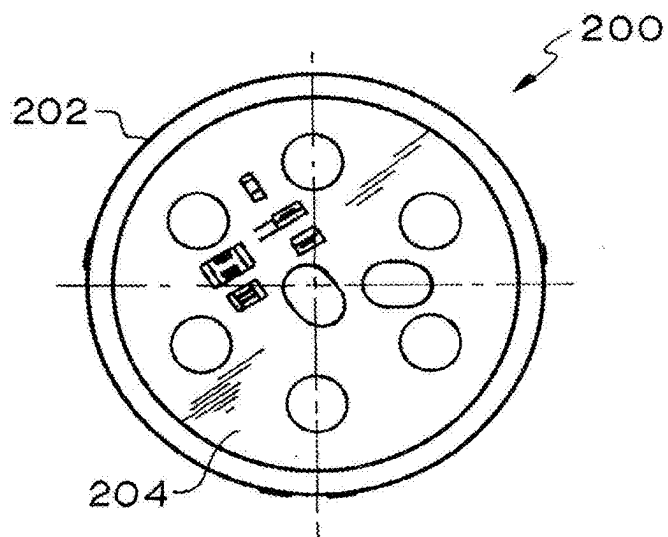


FIG. 2E

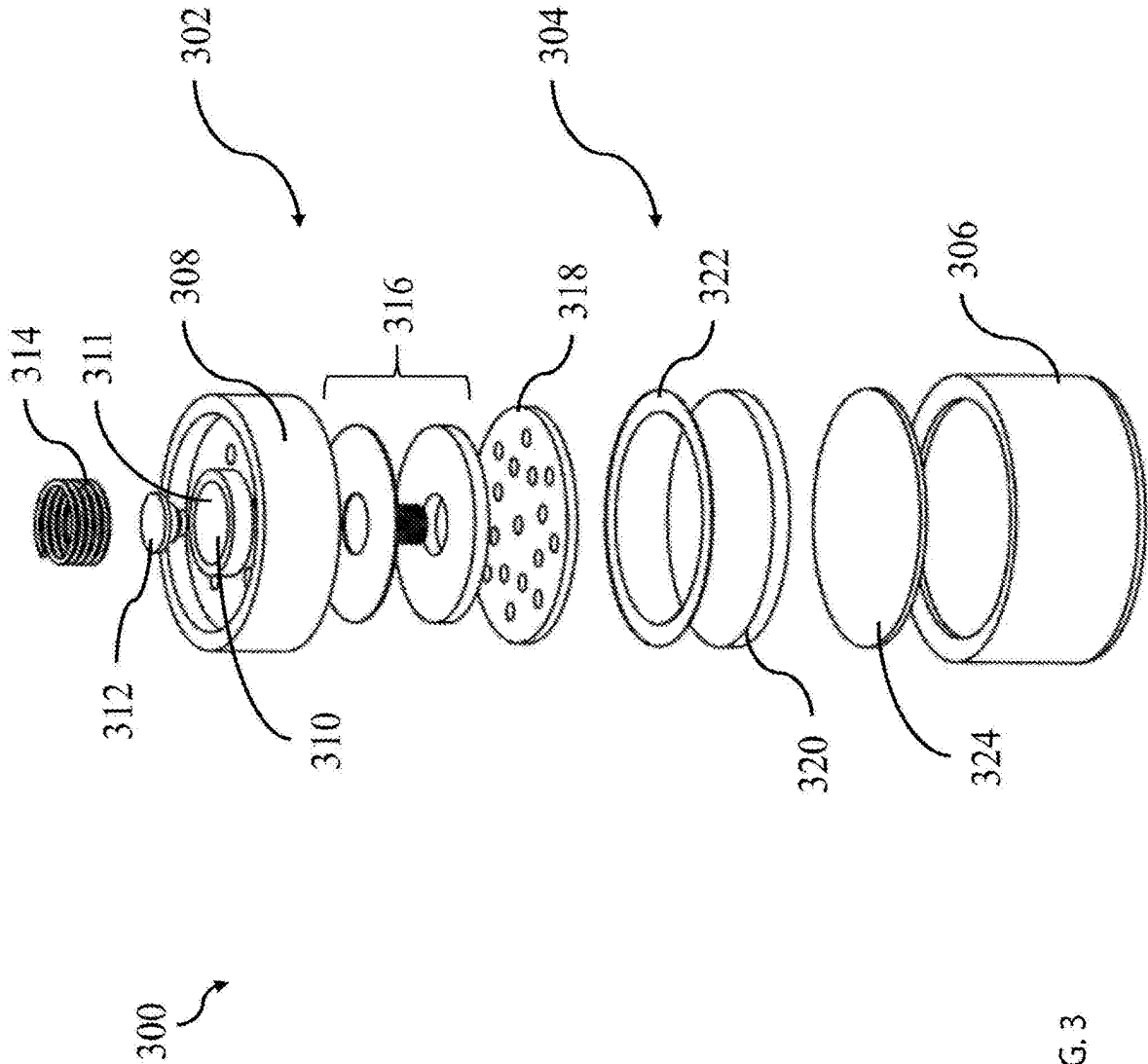


FIG. 3

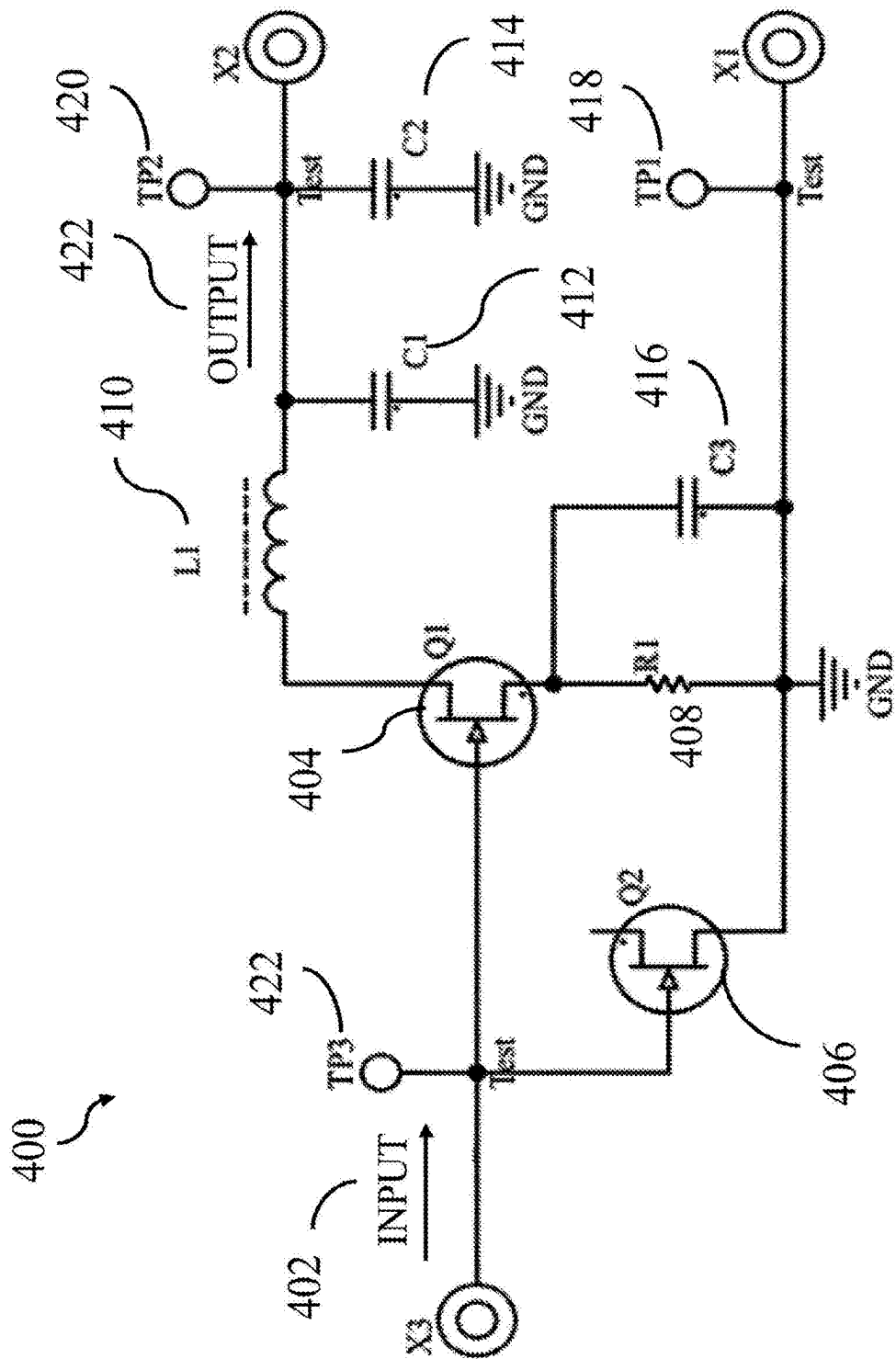


FIG. 4

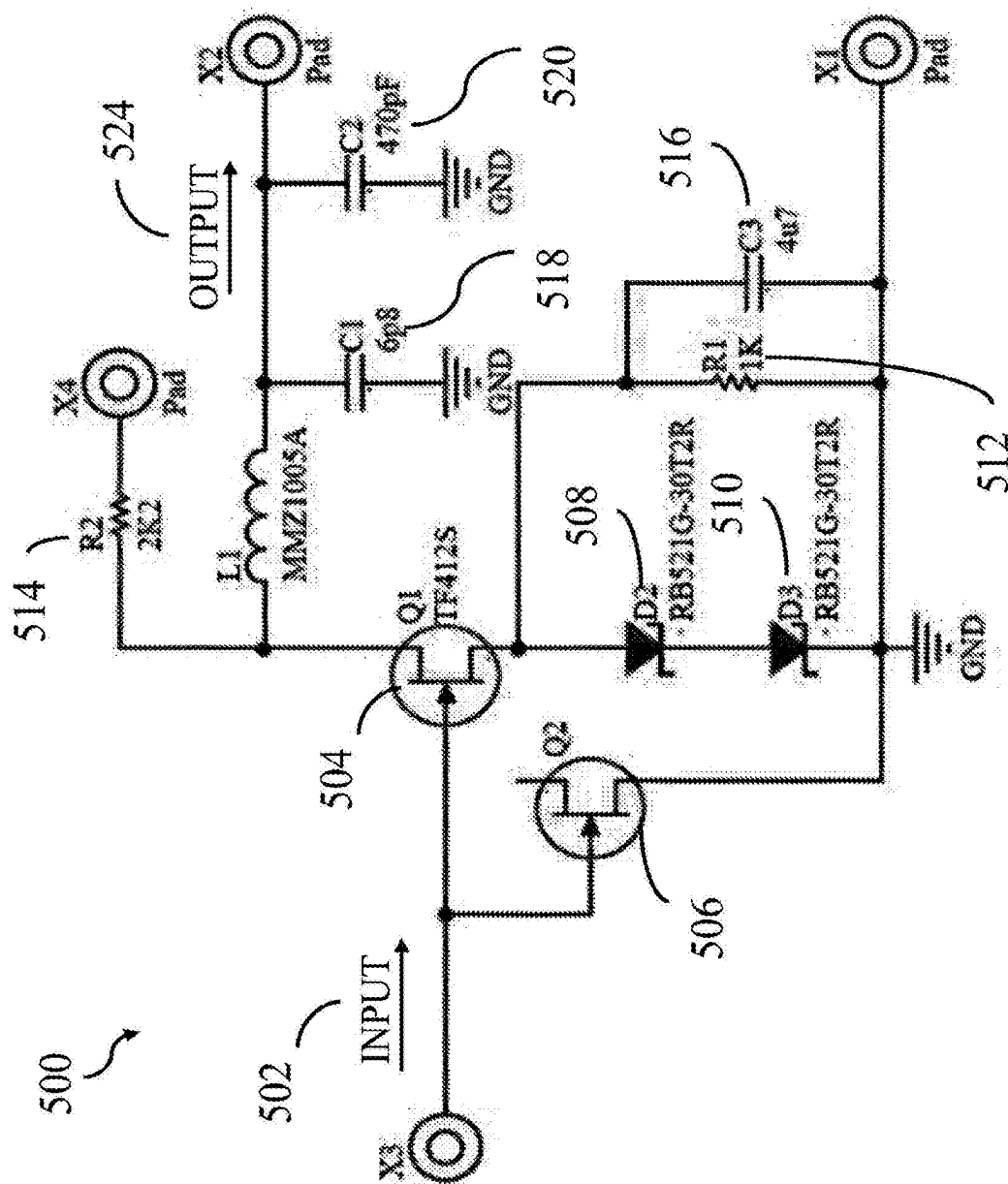


FIG. 5



## ELECTRET CAPSULE CIRCUIT

### CROSS-REFERENCE TO RELATED APPLICATIONS

**[0001]** This application is a divisional of U.S. application Ser. No. 18/044,456, filed Mar. 8, 2023, which is a National Stage Application of PCT/AU2021/051101, Filed Sep. 21, 2021, which claims benefit of Australian Application No. AU2020903377, filed Sep. 21, 2020, each of which is incorporated by reference in its entirety.

### FIELD OF THE INVENTION

**[0002]** The present invention relates generally to the field of electret capsules, and more particularly to a circuit configuration of an impedance converter integrated in an electret capsule such as for use in pre-polarised condenser microphones.

### BACKGROUND

**[0003]** Electret capsules are commonly used within microphone designs and are readily available. However, with the development of micro-electromechanical systems (MEMS) capsules, electret capsules have started to fall out of favor. MEMS capsules are cheaper to produce, and so have replaced electret capsules in a majority of their applications. MEMS capsules, however, suffer from greater noise and are not readily available with directional polar response. Although the effect is insufficient to be a problem for consumer electronics, the signal output quality of high-end microphones such as those used by professionals suffer noticeably.

**[0004]** With the growing popularity of MEMS capsules the supply of electronic components to create low-noise high performance electret capsules has rapidly decreased.

**[0005]** Therefore, there is a need for an alternative approach to electret capsule circuitry that is suitable for use in high-end microphones and provide either equivalent or improved functionality over traditional prior art electret capsules. The present invention satisfies this need.

### SUMMARY

**[0006]** The present invention relates generally to the field of electret capsules, and more particularly to a circuit configuration of an impedance converter integrated in an electret capsule such as for use in condenser microphones.

**[0007]** As detailed below, an electrical circuit of the electret capsule may comprise one or more field-effect transistors (FETs), diodes (such as a zener diode), resistors, capacitors, and inductors.

**[0008]** In one aspect the present invention provides an electret capsule for use in a capsule microphone, comprising one or more field-effect transistors (FETs), biasing diodes, and biasing resistors arranged within an electrical circuit.

**[0009]** In another aspect, the positioning of the biasing diodes and biasing resistors within the electrical circuit may be configured to provide a combined “hybrid bias” upon the FET. Whereas diode biasing provides excellent noise reduction and signal gain but lacks in temperature stability, and resistor biasing provides temperature stability but does not provide any improvement on noise reduction, it is considered that a hybrid biasing configuration provides a synergistic combination of the two prior art biasing forms. In yet another aspect, the electret capsule may further comprise a

“capsule FET” within the electrical circuit, the positioning of which may be configured to bias the FET. In one embodiment, the capsule FET may be in addition to the biasing resistor. In an alternate embodiment, the capsule FET may replace the biasing resistor in the electrical circuit.

**[0010]** In a further aspect, the electret capsule may comprise an inductor within the electrical circuit with one or more capacitors in series. The arrangement of the inductor and capacitors may facilitate improved noise rejection of the circuit with regards to an output signal.

**[0011]** While the invention is susceptible to various modifications and alternative forms, specific exemplary embodiments thereof have been shown by way of example in the drawings and have herein been described in detail. It should be understood, however, that there is no intent to limit the invention to the particular embodiments disclosed, but on the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the scope of the invention as defined by the appended claims.

### BRIEF DESCRIPTION OF THE DRAWINGS

**[0012]** Embodiments are illustrated by way of example and not limitation in the figures in the accompanying drawings, in which like references indicate similar elements and in which:

**[0013]** FIG. 1A illustrates an exploded view of an exemplary microphone assembly including an electret capsule;

**[0014]** FIG. 1B illustrates an exemplary connection of the electret capsule of FIG. 1A;

**[0015]** FIG. 2A illustrates a perspective view of an exemplary electret capsule coupled with an exemplary printed circuit board (PCB);

**[0016]** FIG. 2B illustrates a perspective view of the electret capsule decoupled from the PCB of FIG. 2A;

**[0017]** FIG. 2C illustrates a side view of the electret capsule of FIG. 2A;

**[0018]** FIG. 2D illustrates a bottom view of the electret capsule of FIG. 2A;

**[0019]** FIG. 2E illustrates a top view of the electret capsule and PCB of FIG. 2A;

**[0020]** FIG. 3 illustrates an exploded view of an exemplary electret capsule;

**[0021]** FIG. 4 illustrates an exemplary electret capsule circuit configuration; and

**[0022]** FIG. 5 illustrates another exemplary electret capsule circuit configuration.

### DETAILED DESCRIPTION OF THE INVENTION

**[0023]** The present invention relates generally to the field of electret capsules, and more particularly to a circuit configuration of an impedance converter integrated in an electret capsule such as for use in pre-polarised condenser microphones. The electret capsule of a microphone may include a gate biasing field effect transistor (FET) to facilitate biasing of a low noise FET. Advantageously, the use of a low noise FET in the electret capsule of a microphone provides for improved temperature stability, while achieving lower self-noise. Turning now to the drawings wherein like numerals represent like components, FIG. 1A illustrates an exemplary microphone assembly **100**.

**[0024]** As shown, microphone assembly **100** may include a front cap **102**, a coarse mesh **104**, a fine mesh **106**, an

electret condenser capsule **108**, a wire assembly **110**, a protection mesh **112**, a body **114**, an audio connector **116**, and a rear cap **118**. It is contemplated that any type of microphone or microphone element may be used, such as for example, uni-directional or omni-directional microphones.

[0025] As illustrated, components of microphone **100** may be housed between front cap **102** and rear cap **118** within body **114**. While body **114** is shown to be substantially tubular, other shapes are contemplated. As shown, body **114** may include one or more openings **115** to permit sound waves to enter. Body **114** and front/rear caps **102**, **118** may be made of the same materials or different materials, such as metal, plastic or the like.

[0026] As mentioned above, microphone **100** may include one or more layers of mesh **104**, **106**, **112**. The one or more mesh layers may be used to adjust the acoustic impedance properties of microphone **100** and may prevent intrusion of foreign matter and fine particles. Examples of mesh materials that may be used include nonmetallic (e.g., nonconductive) materials such as woven polyester and PVC-on-polyester fabrics. In general, mesh material may be formed from any suitable fabric material that exhibits acceptable acoustic performance, such as for example, sound transparency of 90% or more.

[0027] As shown in FIGS. 1A-1B, microphone **100** may further include an electret capsule **108** connected to an audio interface **116** via a wire **110**. Electret capsule **108** may include a biasing arrangement of field effect transistors to achieve low noise, as detailed below. While a wire is shown, other connection types are contemplated such as a pin connection or terminal connection. As shown in FIG. 1B, wire **110** may be soldered onto a printed circuit board **120** of the electret capsule **100**.

[0028] Audio connector **116** may facilitate transmitting audio signals, such as analog or digital frequencies to, for example, a speaker. In one instance, audio connector **116** may be an XLR connector having between three and seven pins.

[0029] FIGS. 2A-2E illustrate an exemplary electret capsule **200**. As shown, electret capsule **200** may include a housing **202** and a printed circuit board (PCB) **204**. Housing **202** may include a top surface **206** and a bottom surface **208**. As shown, a PCB **204** may couple with top surface **206**, such as via a contact spring as detailed below.

[0030] Although electret capsule **200** is not limited to specific dimensions, the width of housing **202** may range between about four millimeters and about eight millimeters and preferably between about five millimeters and about seven millimeters. In one embodiment, housing **202** is about six and three tenths millimeters in width. The diameter of electret capsule **200** may range between about four millimeters and about twenty millimeters and preferably between about twelve millimeters and about sixteen millimeters. In one embodiment, the diameter of electret capsule **200** is about fourteen millimeters.

[0031] FIG. 3 illustrates an exploded view of an exemplary electret capsule **300**. As shown, electret capsule **300** may include a backplate assembly **302**, a diaphragm assembly **304**, and a housing **306**. Housing **306** may be configured to hold the components of assemblies **302**, **304**.

[0032] Backplate assembly **302** may include a support member **308**. Support member **308** may include an opening **310** defined by one or more interior walls **311**. An electrically conducting rod **312** may be slidably disposed within

opening **310**. A contact spring **314** is disposed about rod **312** in abutment with the interior walls of opening **310**. Spring **314** may facilitate electrical contact between the rod and damping discs **316**, which may be in contact with a backplate **318**. Damping discs **316** may be constructed of non-conductive porous material such as non-woven fabrics or foams.

[0033] Backplate **318** may be a conductive perforated disc coated in an electret material such as polytetrafluoroethylene (PTFE), which is given a permanent polarization charge. Although not shown in FIG. 3, backplate **318** may be electrically connected to a printed circuit board (PCB) via spring **314**.

[0034] Backplate **318** also may be electrically isolated from a diaphragm **320** of assembly **304** via a non-conductive gasket **322**. Diaphragm **320** may be constructed of a flexible material that is configured to vibrate in response to sound waves. For example, diaphragm **320** may be a flexible film forming a vibratile diaphragm which will respond to sound waves passing through a mesh **324**. Diaphragm **320** may be electrostatically polarized such that vibrations in response to sound waves result in an alternating potential being developed between the diaphragm **320** and the backplate **318**. In addition, backplate **318** may include a plurality of apertures, which contribute to the level of volume between the diaphragm **320** and backplate **318**.

[0035] FIGS. 4 and 5 illustrates exemplary circuits **400**, **500** of an electret capsule including one or more field-effect transistors (FETs), biasing diodes, and biasing resistors arranged within an electrical circuit. It is further contemplated that the capsule may be a kind of a capsule which needs a polarization power supply.

[0036] As illustrated in FIG. 4 and FIG. 5, an electret capsule may include a gate biasing field effect transistor (FET) to facilitate biasing of a low noise FET. Advantageously, the use of low noise FET in an electret capsule provides for a reduced cost, while achieving lower self-noise.

[0037] As illustrated in FIG. 4, exemplary circuit **400** may include two FETs or junction field effect transistor (J-FETs) **404**, **406**, a resistor **408**, and inductor **410**, three capacitors **412**, **414**, **416**, and three test points **418**, **420**, **422**. Each FET **404**, **406** includes a source, a drain, and a gate electrode.

[0038] As shown in FIG. 4, FETs **404**, **406** may be arranged according to a biasing arrangement such that first FET **404** may be an impedance converter low noise FET and said FET **406** may be a gate biasing FET. An input **402** is provided to the gate electrodes of the first FET **404** and second FET **406**. The source electrode of the first FET **404** is connected to source electrode of the second FET **406** and ground through resistor **408**. Capacitor **416** may be configured as a bypass capacitor in parallel with a current setting element, such as resistor **408**.

[0039] The drain electrode of first FET **404** may be connected to ground by inductor **410** and one or more bypass capacitors **412**, **414** in series. The arrangement of the inductor **410** and capacitors **412**, **414** may facilitate improved noise rejection of the circuit with regards to an output signal **422**. Since second FET **406** is used for biasing first FET **404**, the drain electrode of second FET **406** is open.

[0040] As illustrated in FIG. 5, exemplary circuit **500** may include two FETs **504**, **506**, two voltage reference diodes (e.g., zener diodes) **508**, **510**, two resistors **512**, **514**, three

capacitors **516**, **518**, **520**, and an inductor **522**. Each FET **504**, **506** (which may be a J-FET) includes a source, a drain, and a gate electrode.

[0041] As shown, FETs **504**, **506** may be arranged according to a biasing arrangement such that first FET **504** may be an impedance converter low noise FET and said FET **506** may be a gate biasing FET. An input signal **502** may be provided to the gate electrodes of the first FET **504** and second FET **506**.

[0042] The source electrode of the first FET **504** is connected to the source electrode of second FET **506** and ground through two zener diodes **508**, **510** such that a positive bias potential is established at the source. In one embodiment, zener diodes **508**, **510** may be a schottky diode.

[0043] Zener diodes **508**, **510** may be connected in parallel with resistor **512** and capacitor **516**. The positioning of the biasing diodes and biasing resistors within the electrical circuit may be configured to provide a combined “hybrid bias” upon the FETs **504**, **506**. While resistor **512** is shown to have a resistance of 1K, the value may be varied so as to compensate for variations in the characteristics of FETs **504**, **506**.

[0044] The drain electrode of first FET **504** may be connected to ground by inductor **522** and one or more bypass capacitors **518**, **522** in series. The arrangement of the inductor **522** and capacitors **518**, **520** may facilitate improved noise rejection of the circuit with regards to an output signal **524**.

[0045] The drain electrode of first FET **504** may also be connected in series with resistor **514**, which may be configured as a current limiting resistor. Since second FET **506** is used for biasing first FET **504**, the drain electrode of second FET **506** is open.

[0046] As illustrated in FIG. 4 and FIG. 5, a gate biasing FET or J-FET may facilitate the biasing of a low noise FET or J-FET, thereby avoiding the need for cost and size prohibitive components otherwise required to achieve said biasing. Doing so enables the use of low noise J-FET in an electret capsule microphone at a reduced cost, while achieving lower self-noise.

[0047] Further modifications and alternative embodiments of various aspects of the invention will be apparent to those skilled in the art in view of this description. Accordingly, this description is to be construed as illustrative only and is for the purpose of teaching those skilled in the art the general manner of carrying out the invention. It is to be understood that the forms of the invention shown and described in the application are to be taken as examples of embodiments. Components may be substituted for those illustrated and described in the application, parts and processes may be

reversed, and certain features of the invention may be utilized independently, all as would be apparent to one skilled in the art after having the benefit of this description of the invention. Changes may be made in the elements described in the application without departing from the spirit and scope of the invention as described in the following claims.

1. An electret capsule circuit comprising:

a first and a second field effect transistor (FET), each FET including a source, a drain, and a gate;

a biasing arrangement of said first and second FET, wherein said first FET is an impedance converter low noise FET and said second FET is a gate biasing FET, a source electrode of the first FET being electrically connected to a source electrode of the second of FET; a current setting element coupled to the source of said first FET; and

a bypass capacitor in parallel with said current setting element.

2. The electret capsule circuit of claim 1, wherein said current setting element is a biasing resistor including a first terminal connected in series to said source electrode of said first FET and a second terminal connected to ground.

3. The electret capsule circuit of claim 2, wherein said resistor has a resistance value of about 1K $\Omega$  to facilitate compensating for variations in output characteristics of said two or more FETs.

4. The electret capsule circuit of claim 1, wherein said first FET and said second FET are junction field effect transistors (J-FETs).

5. The electret capsule circuit of claim 1, wherein the drain of said second FET is open.

6. The electret capsule circuit of claim 1, wherein said current setting element is a biasing diode configured to reduce noise and signal gain, said diodes connected in series and arranged to connect said first FET and said second FET to ground to facilitate a positive bias potential.

7. The electret capsule circuit of claim 6, wherein said biasing diodes are at least one of a zener diode or a schottky diode.

8. The electret capsule circuit of claim 1, wherein the drain of said first FET is connected to ground by a current limiting resistor.

9. The electret capsule of claim 1, wherein the drain of said first FET is connected to ground by an inductor in series with one or more bypass capacitors in parallel.

10. The electret capsule of claim 9, wherein the drain of said first FET is connected in series with a current limiting resistor.

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